

Contemporary Issues in Intellectual Property Law - Right to Repair

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Topics to be covered:

1. What the right to repair is and why it is important
2. Industry Examples of Repair Restrictions
3. How manufacturers enforce repair monopolies
4. Arguments against the Right to Repair
5. Global Development regarding Repair Laws by country

Right to Repair: Meaning & Importance

- Right to Repair (RtR): ability of owners to repair products they purchase
- Includes access to spare parts, tools, manuals, and diagnostic software
- Challenges manufacturer control after sale
- Key question: what does ownership mean in a digital economy?

Setting the Context

- IP laws originally promoted innovation
- Increasingly used to control repair and modification
- Shift from mechanical goods to software-dependent products
- Repair now requires manufacturer permission
- Embedded software is essential for product function
- Without software, the product cannot operate as intended
- Licensing agreements replace traditional ownership
- Software locks can prevent repair even when hardware works

Why Repair Matters

- Environmental sustainability
- Consumer rights and competition
- Innovation and learning
- Access to repair knowledge
- Social and economic justice





Sustainability

- Repair extends product lifespan
- Reduces extraction of raw materials
- Limits electronic and mechanical waste
- Restrictions encourage premature replacement
- Smartphones & consumer electronics
– repair restrictions accelerate replacement → growing e-waste



Consumer Rights

- Repair is closely linked to ownership
- Restrictions raise repair costs
- Independent repair markets eliminated
- Manufacturers retain control post-sale
- *ACCC v Apple (2018) FCA 953* – Apple misled consumers that third-party repair voided their legal rights

Innovation



- Repair encourages experimentation and tinkering
- Independent repairers identify design flaws
- Bottom-up innovation is restricted by repair bans
- *Apple Inc. v. Corellium, LLC*, 63 F.4th 1353 (11th Cir. 2023) – copyright claims used to challenge independent security research



Repair Knowledge

- Repair involves diagnosis, implementation, and reporting
- Manuals, schematics, and software are essential
- Knowledge often classified as proprietary IP
- Restrictions limit learning and research
- Toshiba repair manuals removed in Australia after IP claims

Social & Economic Justice



- Disproportionate impact on farmers and small businesses
- Serious consequences for hospitals and disabled users
- Low-income consumers rely on affordable repair
- Repair access promotes equity
- Healthcare: *Italy COVID-19 ventilator valves* – repair blocked by IP threats
- Agriculture: *Farmers face harvest delays due to John Deere software locks*

Industry Examples of Repair Restrictions

- Consumer electronics and IoT devices
 - **Google Nest** – Hardware remains functional, but withdrawal of software/cloud support renders devices unusable
 - **Sonos** – “Recycle mode” permanently disables working devices via software command
- Healthcare and medical equipment
 - **Ventilators / infusion pumps** – Repairs require proprietary software and manufacturer authorisation, delaying critical fixes
- Agricultural machinery and vehicles
 - **John Deere tractors** – Proprietary diagnostic software and DRM prevent farmers from completing repairs independently

How Manufacturers Enforce Repair Monopolies

- **Legal barriers: copyright, patents, trade secrets**
 - *Copyright law* restricts access to repair manuals, diagnostic software, and firmware, even when modification is necessary for lawful repair
 - *Anti-circumvention rules* criminalise bypassing digital locks, regardless of whether the purpose is repair or maintenance
 - *Patent law* grants exclusive rights over spare parts and diagnostic tools, preventing third-party manufacture of compatible components
 - *Trade secret* claims are used to withhold schematics, service manuals, and calibration data by classifying them as proprietary information

- **Technological barriers: DRM and digital locks**

- *Digital Rights Management (DRM)* prevents access to embedded software required for diagnostics, calibration, or part replacement
- *Software authentication* systems block the use of non-authorised parts or repairs, even when physically compatible
- *Remote disabling (“bricking”)* allows manufacturers to deactivate devices via software updates or service commands
- *Cloud dependency* ties product functionality to manufacturer-controlled servers; withdrawal of support renders hardware unusable

- **Contractual barriers: EULAs and warranty restrictions**

- *End-User License Agreements (EULAs)* prohibit unauthorised repair, modification, or use of third-party components
- *Warranty restrictions* discourage repair by threatening loss of coverage if independent repair is attempted

Design & Market Strategies

- **Planned obsolescence**

- Products are intentionally designed with shortened lifespans to encourage replacement over repair
- Software updates and support withdrawal render otherwise functional hardware obsolete
- Design choices prioritise upgrades and new models over durability and long-term usability

- **Sealed components and proprietary fasteners**

- Use of glue, welding, and sealed casings prevents non-destructive disassembly
- Proprietary screws and connectors require specialised tools unavailable to consumers
- Key components (e.g. batteries, displays) cannot be replaced independently

- **Restricted access to spare parts**

- Manufacturers withhold parts from consumers and independent repairers
- Exclusive supply arrangements favour authorised service providers
- Parts are bundled or discontinued, forcing full-unit replacement

- **Remote disabling of devices**

- Software commands can permanently deactivate devices (“bricking”)
- Functionality can be limited or withdrawn remotely after sale
- Product end-of-life is determined digitally rather than by physical failure

Arguments Against the Right to Repair

- Safety risks (e.g., batteries, emissions)
- Cybersecurity and data protection concerns
- Protection of intellectual property
- Risk of regulatory non-compliance

Critical Responses

- Independent repairers already operate safely
- Little evidence of increased security risks
- Monopoly control drives many restrictions
- Consumer choice is undermined

Context of the Right to Repair Movement

- Response to repair restrictions and monopolies
- Driven by consumer, environmental, and justice concerns
- Supported by grassroots repair communities
- Increasing political and legal attention

Global Developments in Repair Laws

- **United States: focus on consumer autonomy**

- **Motor Vehicle Owner's Right to Repair Act (2001)** – first statutory recognition of repair access
- **Massachusetts Right to Repair Law (2012; expanded 2020)** – mandates access to vehicle diagnostics and telematics
- **New York Digital Fair Repair Act (2022)** – requires manufacturers to provide parts, tools, and manuals for electronics
- Ongoing federal proposals: **REPAIR Act**, **Fair Repair Act** (not yet enacted)

- **European Union: sustainability and repair-by-design**

- **Ecodesign Directive (2009/125/EC)** – mandates repairability and spare-part availability for certain products
- **Circular Economy Action Plan (2020)** – promotes product longevity and repair over replacement
- **Proposed Right to Repair Directive (2023)** – strengthens consumer repair rights post-sale

- **United Kingdom – Repair-by-Design Approach**

- Ecodesign for Energy-Related Products and Energy Information Regulations 2021

- **Australia**

- 2021 Productivity Commission Right to Repair Report

- **Canada**

- Copyright focused reforms

- **France**

- Anti-Waste for a Circular Economy Law (2020):
Introduced a mandatory repairability index (score out of 10) for electronic products
- Energy Transition for Green Growth Law: Criminalises planned obsolescence

Thank you